

Disturbance Rejection Control for a Ducted-Fan UAV Using Priority-Enhanced INDI

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Abstract. The Ducted-Fan UAV is susceptible to external disturbances during flight. Meanwhile, its inner-loop angular-velocity dynamics constitute an input-constrained over-actuated system, which couples disturbance rejection with control allocation. To address these issues, this paper proposes a Priority-enhanced INDI (PINDI) control framework for the Ducted-Fan UAV. In the translational outer loop, an INDI-based acceleration controller is introduced to reduce the dependence on explicit aerodynamic modeling. In the attitude-feedback loop, a canonicalized quaternion attitude error is adopted to provide a consistent attitude-error representation. For the input-constrained over-actuated angular-velocity dynamics, the PINDI inner loop decomposes the virtual control command into a trim-control term and an error-feedback term, and introduces a scaling factor for direction-preserving allocation under actuator constraints. In this way, disturbance rejection and control allocation are handled within a unified framework, improving the attitude stability and disturbance-rejection performance of the system under sustained external disturbances. The proposed method is validated on a high-fidelity flight simulation platform, and the results demonstrate that it can maintain stable flight and achieve reliable control allocation under sustained external disturbances.

Keywords: Disturbance Rejection Control, Ducted-Fan UAV, PINDI, Fixed-Point Hovering, Control Allocation

1 Introduction

In recent years, unmanned aerial vehicles (UAVs) have been increasingly used in complex-environment operations, close-range hovering, and high-precision autonomous flight missions, placing growing demands on disturbance rejection and dynamic performance. Although multicopter control technology has advanced rapidly, the stability, robustness, and engineering realizability of flight controllers

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under external disturbances, model uncertainties, and demanding flight conditions still require further study [1]. PID remains widely used for its simple structure and ease of implementation, but its performance is often limited under strong nonlinearities and sustained disturbances. Geometric control defines attitude errors directly on $SO(3)$, avoiding Euler-angle singularities and showing effectiveness in aggressive maneuver tracking [2]. Incremental Nonlinear Dynamic Inversion (INDI) constructs control laws from local incremental input–output relationships, thereby reducing dependence on model accuracy. Recent studies have systematically summarized the development and applications of INDI-type methods in flight control [3].

For ducted-fan UAVs, the control problem is further complicated by the duct-induced coupling between the rotor and the surrounding flow field. Although the ducted configuration provides favorable safety characteristics and potential aerodynamic efficiency [4], additional aerodynamic forces and moments may arise in low-speed, hovering, and confined-space conditions [5]. Recent studies on ducted-fan tail-sitter UAVs have shown the complexity of altitude-hold mode transition under strong aerodynamic coupling [6,7]. In addition, Shan *et al.* combined INDI with priority-based control allocation for attitude control of a ducted-fan UAV, emphasizing the need to preserve the feasibility of the INDI linearization term under input redundancy and control-surface constraints [8]. Along this line, Priority-enhanced INDI (PINDI) incorporates priority-based control allocation into the INDI framework to handle actuator constraints, with the key objective of preserving the direction of the desired incremental control effect through suitable scaling. Meng *et al.* further analyzed the stability and robustness of this method for over-actuated systems [9], while related studies on control allocation for fully actuated aerial vehicles provide useful background for the present work [10].

In the presence of complex aerodynamic coupling effects, this paper proposes a PINDI-based flight-control framework for hovering control of a ducted-fan UAV under disturbance. The framework retains the cascaded control architecture and incorporates PINDI into the inner loop to combine angular-velocity regulation, disturbance compensation, and actuator-constrained control allocation. The main contributions of this paper are as follows:

1. An INDI-based acceleration controller is introduced into the translational loop to reduce the dependence on explicit aerodynamic modeling, while a PINDI angular-velocity controller is developed for the over-actuated inner-loop dynamics by introducing a scaling factor for direction-preserving allocation under actuator constraints.
2. A canonicalized quaternion attitude error is adopted to alleviate the sign ambiguity of unit quaternions and provide a locally consistent feedback variable.
3. PX4/Gazebo comparative simulations are conducted to validate the proposed controller against the baseline PX4 cascaded PID controller.³

The remainder of this paper is organized as follows. Section 2 presents the controller-oriented modeling of the ducted-fan UAV. Section 3 details the pro-

³ <https://github.com/mengchaoheng/DuctedFanUAV-Autopilot>

posed cascaded control design, including position and velocity control, acceleration control, attitude control, angular-velocity control, and direction-preserving allocation. Section 4 presents the simulation setup and comparative results. Finally, Section 5 concludes the paper.

2 Modeling

This section presents the controller-oriented model of the ducted-fan UAV, including the coordinate frames, 6-DoF rigid-body dynamics, and control-surface moment mapping, as illustrated in Fig. 1.

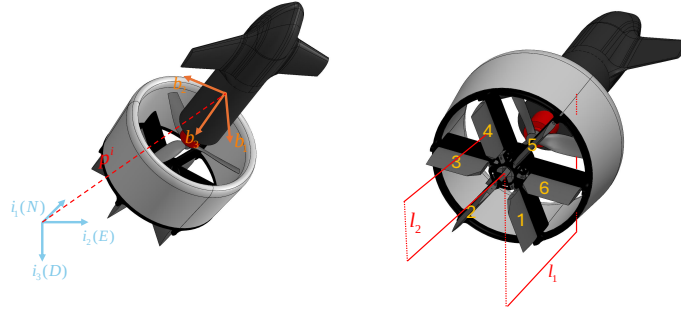


Fig. 1. Aircraft configuration, coordinate frames, and control-surface layout.

The inertial frame $\{\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3\}$ is defined as a North-East-Down (NED) frame, with its origin fixed at an arbitrary point on the Earth. The body-fixed frame $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ is attached to the aircraft, with its origin located at the center of mass. $\mathbf{R}_b^i = [\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3] \in SO(3)$ represents the transformation from the body frame to the inertial frame, where $SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}\mathbf{R}^\top = \mathbf{I}, \det(\mathbf{R}) = 1\}$.

The position and velocity of the body frame relative to the inertial frame are denoted by $\mathbf{p}^i = [x \ y \ z]^\top$ and $\mathbf{v}^i = [v_x \ v_y \ v_z]^\top$, respectively. The attitude is described using the ZYX rotation sequence, corresponding to the yaw angle ψ , pitch angle θ , and roll angle φ . The 6-DoF motion equations of the aircraft are given by

$$\begin{cases} \dot{\mathbf{p}}^i = \mathbf{v}^i \\ \dot{\mathbf{v}}^i = m^{-1} \mathbf{R}_b^i \mathbf{F}^b + g \mathbf{i}_3 \\ \dot{\mathbf{R}}_b^i = \mathbf{R}_b^i [\boldsymbol{\omega}^b]_\times \\ \dot{\boldsymbol{\omega}}^b = \mathbf{J}^{-1} (\mathbf{M}^b - \boldsymbol{\omega}^b \times \mathbf{J} \boldsymbol{\omega}^b) \end{cases}, \quad (1)$$

where g is the gravitational constant, and m is the mass of the aircraft. $\mathbf{J} = \text{diag}(J_x, J_y, J_z) \in \mathbb{R}^{3 \times 3}$ denotes the inertia matrix and $\boldsymbol{\omega}^b = [p, q, r]^\top$ is the angular-velocity vector, whose components p , q , and r are the roll, pitch, and yaw rates, respectively.

The total non-gravitational force and moment acting on the aircraft are decomposed as

$$\mathbf{F}^b = \mathbf{F}_T^b + \mathbf{F}_a^b, \quad \mathbf{M}^b = \mathbf{M}_c^b + \mathbf{M}_a^b, \quad (2)$$

where $\mathbf{F}_T^b$ denotes the ducted-fan thrust force, $\mathbf{F}_a^b$ collects the remaining aerodynamic forces, $\mathbf{M}_c^b$ is the control moment generated by the control surfaces, and $\mathbf{M}_a^b$ denotes the remaining aerodynamic moments.

The thrust force is modeled as

$$\mathbf{F}_T^b = [0, 0, -T]^\top, \quad T = k_T \Omega^2, \quad (3)$$

where Ω is the rotor speed, and k_T is the thrust coefficient.

The vehicle considered in this paper is a self-developed ducted-fan UAV named SHC09, equipped with six control surfaces uniformly distributed at an angular spacing of 60° . Deflections of these control surfaces generate aerodynamic effects in both the force and moment channels. For controller design, the control-surface input is modeled through the moment term $\mathbf{M}_c^b$, while the residual aerodynamic force and moment are included in $\mathbf{F}_a^b$ and $\mathbf{M}_a^b$, respectively. The vector of control-surface deflections is denoted by $\mathbf{u} \in \mathbb{R}^6$.

$$\mathbf{M}_c^b = \frac{1}{2} k_{cv} \begin{bmatrix} -l_1 & -\frac{1}{2}l_1 & \frac{1}{2}l_1 & l_1 & \frac{1}{2}l_1 & -\frac{1}{2}l_1 \\ 0 & \frac{\sqrt{3}}{2}l_1 & \frac{\sqrt{3}}{2}l_1 & 0 & -\frac{\sqrt{3}}{2}l_1 & -\frac{\sqrt{3}}{2}l_1 \\ l_2 & l_2 & l_2 & l_2 & l_2 & l_2 \end{bmatrix} \mathbf{u}, \quad (4)$$

where k_{cv} is the lift coefficient of the control surfaces, and l_1 and l_2 denote the distances from the lift application point to the rotation center.

3 Controller Design

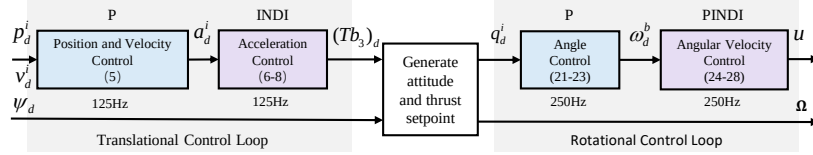


Fig. 2. Cascaded flight-control architecture.

As shown in Fig. 2, the proposed controller adopts a cascaded architecture with an INDI-based acceleration outer loop and a PINDI-based angular-velocity inner loop. The former computes the thrust command with reduced reliance on explicit aerodynamic modeling, while the latter handles angular-velocity regulation and actuator-constrained control allocation.

3.1 Position and Velocity Control

The desired acceleration is computed by a position and velocity proportional controller as

$$\mathbf{a}_d^i = \mathbf{K}_p(\mathbf{p}_d^i - \mathbf{p}^i) + \mathbf{K}_v(\mathbf{v}_d^i - \mathbf{v}^i), \quad (5)$$

where $\mathbf{K}_p$ and $\mathbf{K}_v$ are positive-definite gain matrices, and $\mathbf{p}_d^i$ and $\mathbf{v}_d^i$ denote the desired position and velocity in the inertial frame, respectively.

From (1), (2), and (3), the translational dynamics are driven by the thrust force, whose inertial-frame expression is $-T\mathbf{b}_3$. Meanwhile, the aerodynamic force $\mathbf{F}_a^b$ is unknown and may vary significantly under sustained external disturbances. Therefore, instead of explicitly reconstructing the desired force decomposition from (2), we directly design an INDI-based acceleration controller using the incremental relationship between the translational acceleration and the thrust-vector input. In this way, the influence of the unknown aerodynamic force can be compensated in an incremental manner.

3.2 Acceleration Control

This section designs an INDI-based acceleration controller, in which the unknown aerodynamic-force term $\mathbf{F}_a^b$ is naturally embedded in the incremental acceleration response. We treat the inertial-frame thrust vector $T\mathbf{b}_3$ as the virtual control input. Based on (1), the translational dynamics can be rewritten around the previous operating point by a first-order Taylor expansion as

$$\dot{\mathbf{v}}^i = \dot{\mathbf{v}}_0^i - m^{-1} [(T\mathbf{b}_3) - (T\mathbf{b}_3)_0] + \mathcal{O}, \quad (6)$$

where $\dot{\mathbf{v}}_0^i$ and $(T\mathbf{b}_3)_0$ are the acceleration and thrust vector at the previous time step, respectively, and $\mathcal{O}$ represents the higher-order term, which is neglected in the controller design.

The effects of the unknown aerodynamic force and gravity are absorbed into $\dot{\mathbf{v}}_0^i$. Therefore, only the previous translational acceleration needs to be estimated. Using the IMU measurement, the translational acceleration in the inertial frame can be obtained as

$$\mathbf{a}^i = \mathbf{R}_b^i \mathbf{a}_m^b + g\mathbf{i}_3, \quad (7)$$

where $\mathbf{a}_m^b$ is the acceleration measured by the IMU in the body frame. Using the attitude estimate $\mathbf{R}_b^i$ provided by the onboard estimator, the measured body-frame acceleration is transformed into the inertial frame to obtain $\mathbf{a}^i$. Let $\dot{\mathbf{v}}_0^i = \mathbf{a}_f^i$, where $\mathbf{a}_f^i$ is the low-pass filtered value of $\mathbf{a}^i$. Then, the desired thrust vector can be calculated as

$$(T\mathbf{b}_3)_d = (T\mathbf{b}_3)_0 - m (\mathbf{a}_d^i - \dot{\mathbf{v}}_0^i). \quad (8)$$

3.3 Attitude Control

According to the desired thrust vector $(T\mathbf{b}_3)_d$, the desired thrust direction and thrust magnitude are determined by

$$\mathbf{b}_3^d = \frac{(T\mathbf{b}_3)_d}{\|(T\mathbf{b}_3)_d\|}, \quad T_d = \|(T\mathbf{b}_3)_d\|, \quad (9)$$

where $\mathbf{b}_3^d$ and T_d denote the desired thrust direction and desired thrust magnitude, respectively.

Given the reference heading angle ψ_d , an intermediate heading vector in the inertial frame is defined as

$$\mathbf{y}_{ref} = [-\sin \psi_d, \cos \psi_d, 0]^\top. \quad (10)$$

Then, the desired body axes can be constructed as

$$\mathbf{b}_1^d = \frac{\mathbf{y}_{ref} \times \mathbf{b}_3^d}{\|\mathbf{y}_{ref} \times \mathbf{b}_3^d\|}, \quad \mathbf{b}_2^d = \mathbf{b}_3^d \times \mathbf{b}_1^d. \quad (11)$$

Accordingly, the desired attitude matrix is obtained as

$$\mathbf{R}_d^i = [\mathbf{b}_1^d \ \mathbf{b}_2^d \ \mathbf{b}_3^d]. \quad (12)$$

The attitude error can be described using different representations of the same relative rotation. To formulate the quaternion-based feedback variable adopted in this paper, the unit-quaternion representation is introduced as follows. A unit quaternion associated with the axis-angle representation can be written as

$$\mathbf{q} = \begin{bmatrix} q_1 \\ \mathbf{q}_v \end{bmatrix} = \begin{bmatrix} \cos(\phi/2) \\ \mathbf{n} \sin(\phi/2) \end{bmatrix} \in S^3, \quad (13)$$

where q_1 and $\mathbf{q}_v \in \mathbb{R}^3$ are the scalar and vector parts, respectively. The corresponding rotation vector is denoted by $\boldsymbol{\phi} = \phi \mathbf{n}$, where ϕ is the rotation angle and $\mathbf{n} \in \mathbb{R}^3$ is the unit rotation axis.

Let $\mathbf{q}$ and $\mathbf{q}_d$ denote the unit quaternions corresponding to the current attitude and the desired attitude, respectively. We define the attitude error as

$$\mathbf{q}_e = \mathbf{q}^{-1} \otimes \mathbf{q}_d, \quad \mathbf{R}_e = (\mathbf{R}_b^i)^\top \mathbf{R}_d^i. \quad (14)$$

where $\mathbf{q}_e = [q_{e,1} \ \mathbf{q}_{e,v}^\top]^\top \in S^3$ and $\mathbf{R}_e \in SO(3)$ denote the attitude error represented in quaternion form and rotation matrix form, respectively. According to the logarithmic-map formulation in [11], the corresponding rotation vector in $\mathbb{R}^3$ can be written as

$$\boldsymbol{\phi}_e = \phi_e \mathbf{n}_e = \text{Log}(\mathbf{R}_e) = \text{Log}(\mathbf{q}_e), \quad (15)$$

$$\phi_e = \arccos\left(\frac{\text{trace}(\mathbf{R}_e) - 1}{2}\right), \quad \mathbf{n}_e = \frac{(\mathbf{R}_e - \mathbf{R}_e^\top)^\vee}{2 \sin \phi_e}. \quad (16)$$

$$\phi_e = 2 \text{atan2}(\|\mathbf{q}_{e,v}\|, q_{e,1}), \quad \mathbf{n}_e = \frac{\mathbf{q}_{e,v}}{\|\mathbf{q}_{e,v}\|}. \quad (17)$$

Many attitude-error definitions are mathematically related, since they can be regarded as different functions of the same rotation vector in (15). However, their numerical properties are not identical. In particular, some definitions may become ill-defined or degenerate at specific attitudes, which affects their practical use in feedback-controller implementation. To remove the sign ambiguity of the unit quaternion $\mathbf{q}_e$, we define its canonical form as

$$\hat{\mathbf{q}}_e = \text{sign}(q_{e,i^\circ}) \mathbf{q}_e, \quad (18)$$

where

$$\text{sign}(x) = \begin{cases} 1, & x \geq 0, \\ -1, & x < 0, \end{cases} \quad (19)$$

$$i^\circ = \min\{i \in \{1, 2, 3, 4\} \mid q_{e,i} \neq 0\}. \quad (20)$$

Accordingly, the quaternion-based attitude error vector adopted in this paper is defined as

$$\mathbf{e}_q = 2\hat{\mathbf{q}}_{e,v} = 2\text{sign}(q_{e,i^\circ})\mathbf{q}_{e,v}. \quad (21)$$

The above definition provides a vector-form attitude error directly in $\mathbb{R}^3$ and preserves a clear geometric relation with the underlying axis-angle parameterization. Moreover, the sign correction alleviates the ambiguity caused by the double-cover property of unit quaternions, so that the error variable remains locally consistent around the reference attitude. For small attitude errors, it further satisfies

$$\mathbf{e}_q = 2\sin(\phi_e/2)\mathbf{n}_e \approx \phi_e\mathbf{n}_e, \quad (22)$$

which is convenient for controller design and the subsequent Lyapunov-based stability analysis.

Representative attitude-error definitions are compared in Fig. 3. It can be seen that, although these definitions are mathematically related, their norm characteristics near critical attitudes are different, reflecting the influence of their numerical singularities and degeneracy properties.

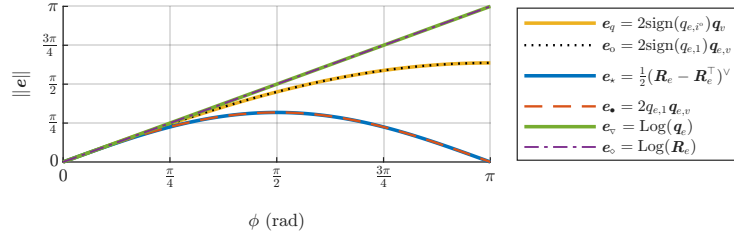


Fig. 3. Comparison of the norms of representative attitude-error definitions.

The desired body angular velocity is designed as

$$\boldsymbol{\omega}_d^b = \mathbf{K}_q \mathbf{e}_q, \quad (23)$$

where $\mathbf{K}_q \in \mathbb{R}^{3 \times 3}$ is the attitude-loop proportional gain matrix.

3.4 Angular Velocity Control

The angular-velocity controller is organized in a modular form. Rewriting the angular-velocity dynamics in (1) yields

$$\dot{\boldsymbol{\omega}}^b = \boldsymbol{\nu} + (\dot{\boldsymbol{\omega}}_0^b - \mathbf{B}\mathbf{u}_0 + \boldsymbol{\delta}), \quad \boldsymbol{\nu} = \mathbf{B}\mathbf{u}, \quad (24)$$

$$\mathbf{B} = \frac{1}{2}k_{cv}\mathbf{J}^{-1}, \begin{bmatrix} -l_1 & -\frac{1}{2}l_1 & \frac{1}{2}l_1 & l_1 & \frac{1}{2}l_1 & -\frac{1}{2}l_1 \\ 0 & \frac{\sqrt{3}}{2}l_1 & \frac{\sqrt{3}}{2}l_1 & 0 & -\frac{\sqrt{3}}{2}l_1 & -\frac{\sqrt{3}}{2}l_1 \\ l_2 & l_2 & l_2 & l_2 & l_2 & l_2 \end{bmatrix} \quad (25)$$

where δ collects the higher-order terms neglected in the incremental linearization, $\dot{\omega}_0^b$ and $\mathbf{u}_0$ denote the angular acceleration and control input at the previous sampling instant, respectively.

Accordingly, the PINDI control law is designed as

$$\boldsymbol{\nu} = \boldsymbol{\nu}_h + \rho \boldsymbol{\nu}_c, \quad \boldsymbol{\nu}_h = \mathbf{B}\mathbf{u}_0 - \dot{\omega}_0^b, \quad \boldsymbol{\nu}_c = \mathbf{K}_\omega(\omega_d^b - \omega^b). \quad (26)$$

where $\boldsymbol{\nu}_h$ is the trim-control term that compensates for system nonlinearities and external disturbances, $\boldsymbol{\nu}_c$ is the error-feedback term responsible for angular-velocity reference tracking [9], and $\mathbf{K}_\omega$ is the corresponding gain matrix. When the required $\boldsymbol{\nu}$ cannot be obtained, a scaling factor $\rho \in [0, 1]$ is introduced, and the control allocation problem is modified to solve for a suitable pair $(\mathbf{u}, \rho)$. In this way, a feasible virtual control command $\boldsymbol{\nu}$ consistent with the control objective can be obtained under the dual requirements of preserving the direction of the error-feedback term $\boldsymbol{\nu}_c$ and satisfying the actuator saturation limits.

Considering the direction-preserving control allocation problem under $\mathbf{u} \in \mathbb{U}$, with $\mathbb{U} = \{\mathbf{u} \in \mathbb{R}^6 \mid u_{\min} \leq u_j \leq u_{\max}, j = 1, \dots, 6\}$, both $\mathbf{u}$ and ρ are treated as decision variables, leading to the following linear programming problem:

$$\begin{aligned} \max_{\rho, \mathbf{u}} \quad & \rho \\ \text{s.t.} \quad & \mathbf{u} \in \mathbb{U}, \\ & 0 \leq \rho \leq 1, \\ & \mathbf{B}\mathbf{u} = \boldsymbol{\nu}_h + \rho \boldsymbol{\nu}_c. \end{aligned} \quad (27)$$

The actual value of ρ is obtained as the solution of the optimization problem. When the desired virtual control command $\boldsymbol{\nu}$ is attainable, ρ is equal to one. When the command exceeds the available control authority, ρ decreases to scale the error-feedback term $\boldsymbol{\nu}_c$, thereby generating an attainable virtual control command $\boldsymbol{\nu}$. This mechanism may reduce the transient response speed, but it can prevent actuator saturation and closed-loop instability caused by unattainable commands.

The direction-preserving stage of the priority-based control-allocation scheme is then implemented in two steps. First, the linear programming problem in (27) is solved. If it is feasible, the allocation process is completed. Otherwise, the error-feedback term is temporarily removed and the equality constraint in (27) is replaced by

$$\mathbf{B}\mathbf{u} = \rho \boldsymbol{\nu}_h, \quad (28)$$

after which the resulting linear programming problem is solved to preserve the most essential control component under the available control authority. The gains are tuned according to the cascaded bandwidth hierarchy, with the angular-velocity loop faster than the attitude loop and the attitude loop faster than the translational loop; practical implementation also requires sufficiently fast sensing, filtering, and actuator response.

4 Simulation Results

We use a software-in-the-loop simulation environment built in Gazebo Classic to verify the effectiveness of the proposed method. The simulation environment is shown in Fig. 4, where the model and controller parameters can be adjusted and external disturbances can be introduced to evaluate robustness.

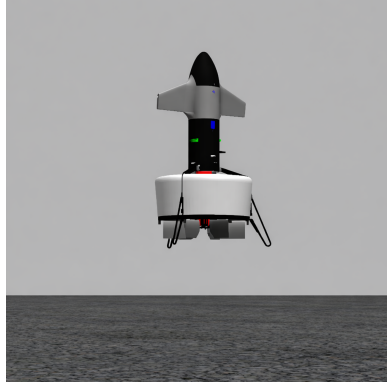


Fig. 4. Simulation environment for hovering comparison under disturbance.

Table 1. Parameters

Parameter	Value	Parameter	Value	Parameter	Value
J_x	0.0438 kg · m ²	J_y	0.0436 kg · m ²	J_z	0.0050 kg · m ²
m	2.100 kg	g	9.788 m/s ²	k_T	2.8964×10^{-5}
l_1	0.267 m	l_2	0.066 m	k_{cv}	0.0032
$\Omega_{\min}$	200	$\Omega_{\max}$	1200	$u_{\min}$	-40°
$u_{\max}$	40°	$\mathbf{K}_p$	diag(1.30, 1.30, 1.00)	$\mathbf{K}_v$	diag(3.20, 3.20, 2.10)
$\mathbf{K}_q$	diag(5.00, 5.00, 4.00)	$\mathbf{K}_\omega$	diag(9.20, 9.20, 7.40)		

The key physical parameters, actuator constraints, and controller gains used in the comparative simulations are summarized in Table 1. The vehicle model, actuator constraints, reference commands, and disturbance settings are kept identical for both controllers, while the controller-specific gains are configured according to their respective control structures.

4.1 Simulation 1: Fixed-Point Hovering Response under Disturbance

To evaluate disturbance rejection, comparative fixed-point hovering simulations are conducted against the baseline PX4 cascaded PID controller. The vehicle is commanded to hover at $\mathbf{p}_d^i = [0, 0, 10]^\top$ m, with identical vehicle model, actuator constraints, reference commands, and disturbance settings for both controllers.

After the ducted-fan UAV reaches the steady hovering condition, an external sinusoidal disturbance moment is applied to the lateral rotational channels of the body frame from $t = t_0$ onward. This setting is used to evaluate the fixed-point

hovering error under sustained moment disturbances.

$$\tau_i(t) = \begin{cases} 0, & t < t_0, \\ c_i + A_i \sin(2\pi f_i(t - t_0)), & t \geq t_0, \end{cases} \quad i \in \{1, 2\}, \quad (29)$$

where A_i , f_i , and c_i denote the amplitude, frequency, and bias term of the disturbance moment in the i -th direction, respectively.

The disturbance-induced deviation from the commanded hovering point is evaluated using the two-norm of the position-error vector

$$\|e\| = \|p_d^i - p^i\|. \quad (30)$$

Here, M_D denotes the disturbance amplitude swept in the experiment, with $A_1 = A_2 = M_D$. With f_i and c_i fixed, increasing M_D strengthens the applied moment disturbance and imposes a higher compensation demand on the closed-loop system.

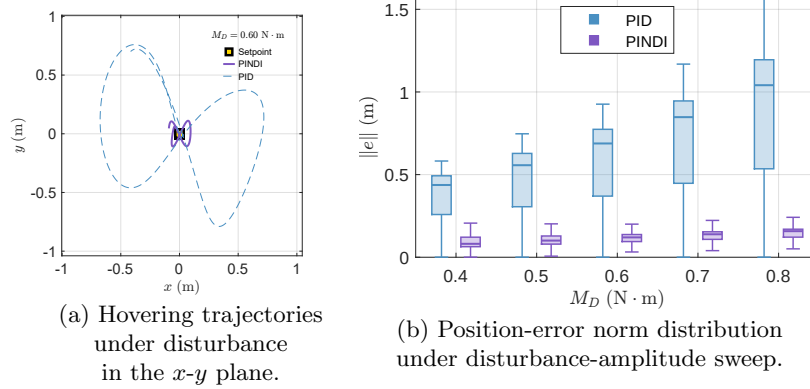


Fig. 5. Fixed-point hovering responses under lateral moment disturbances.

At $M_D = 0.60 \text{ N} \cdot \text{m}$, Fig. 5(a) shows that both controllers exhibit figure-eight-shaped deviations around the commanded hovering point in the x - y plane. This trajectory pattern indicates that the lateral moment disturbance is coupled into the translational motion through the attitude dynamics. The deviation scale obtained with PINDI is visibly smaller than that of PID, suggesting that PINDI can more effectively suppress the disturbance-induced position drift around the reference point.

A further experiment is conducted by increasing M_D from 0.4 to 0.8 $\text{N} \cdot \text{m}$, and the corresponding distributions of $\|e\|$ are summarized in Fig. 5(b). As the disturbance amplitude increases, the PID error distribution shifts upward and becomes more dispersed, indicating a degradation of fixed-point hovering performance under stronger moment disturbances. PINDI maintains lower and more concentrated errors throughout the tested amplitude range, demonstrating better robustness against increasing lateral moment disturbances.

To further explain the position-error reduction in Fig. 5, the translational-velocity and body-rate responses at $M_D = 0.60 \text{ N} \cdot \text{m}$ are compared in Fig. 6.

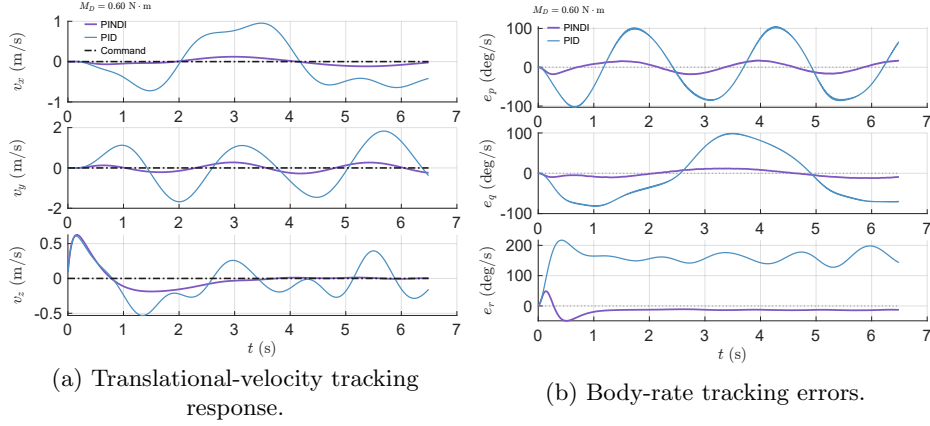


Fig. 6. Translational-velocity and body-rate responses under disturbance.

As shown in Fig. 6, the proposed controller reduces both translational-velocity fluctuations and body-rate tracking errors compared with PID. This result indicates that the INDI outer loop improves translational disturbance compensation, while the PINDI inner loop enhances rotational regulation.

4.2 Simulation 2: Controller-Switching Validation

A controller-switching test is conducted in the same fixed-point hovering scenario at $M_D = 0.60 \text{ N} \cdot \text{m}$.

The ducted-fan UAV is commanded to hover at the same fixed point for 60 s. The baseline PID controller is used during the first 15 s, the proposed PINDI controller is activated from $t = 15 \text{ s}$ to $t = 45 \text{ s}$, and the controller is then switched back to PID during the last 15 s. The position-error norm defined in Eq. (30) is used to evaluate the deviation from the commanded hovering point during the switching process.

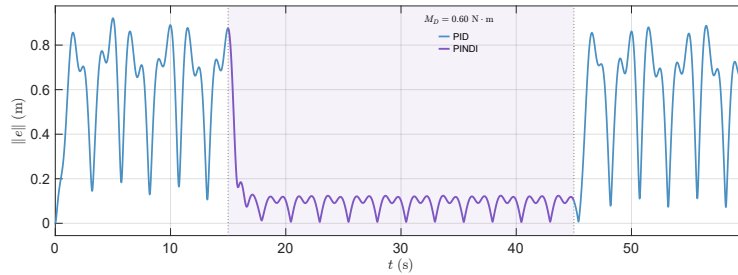


Fig. 7. Position-error norm during controller switching.

As shown in Fig. 7, the error norm decreases after the controller is switched from PID to PINDI and increases again after switching back to PID. This switching behavior directly supports the improved fixed-point hovering capability of the proposed controller under sustained moment disturbances.

5 Conclusion

This paper proposes a priority-enhanced INDI-based flight-control framework for a ducted-fan UAV to improve fixed-point hovering under sustained lateral moment disturbances. The framework combines an INDI outer loop, a canonicalized quaternion attitude-error representation, and a PINDI inner loop with direction-preserving allocation under actuator constraints.

PX4/Gazebo simulations show that the proposed controller achieves a more compact hovering trajectory under disturbance, a smaller position-error norm, and reduced translational-velocity and body-rate deviations compared with the baseline PX4 cascaded PID controller. The disturbance-amplitude sweep and controller-switching test further verify its improved disturbance-rejection capability. Future work will extend the framework to more aggressive flight tasks and more complex disturbance environments.

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